Proof: $\int_0^1 v\mathcal{L}(u)\,dx$ where u & v satisfy BCs

$$\Rightarrow \int_0^1 v\,u^{(iv)}\,dx = -\int_0^1 v'u'''\,dx + [vu''']_0^1$$

$$\Rightarrow \int_0^1 v''u''\,dx + [vu''' - v'u'']_0^1$$

$$\Rightarrow -\int_0^1 v'''u'\,dx + [vu''' - v'u'' + v''u']_0^1$$

$$\Rightarrow \int_0^1 v^{(4)}u\,dx + [vu''' - v'u'' + v''u' - v'''u]_0^1$$

$$\therefore \int_0^1 v\,u^{(4)}\,dx = \int_0^1 u\,v^{(4)}\,dx$$

Primal Problem:

$$\left\{\begin{array}{l} u^{(4)} = f \\ u(0) = u(1) = u'(0) = u'(1) = 0 \end{array}\right\}$$

Dual Problem:

$$\left\{\begin{array}{l} \phi^{(4)} = e \\ \phi(0) = \phi(1) = \phi''(0) = \phi'(1) \end{array}\right\}$$

$$\|e\|^2 = \int_0^1 e^2\,dx = \int_0^1 e\,\phi^{(4)}\,dx = \int_0^1 e''\phi''\,dx$$

Boundary terms vanish: $e(0) = e(1) = 0$

$e'(0) = e'(1) = 0$

Galerkin Orthogonality: $\int_0^1 e''(\pi_h\phi)''\,dx = 0.$

$$\|e\|^2 = \int_0^1 e''(\phi - \pi_h\phi)''\,dx$$

$$\leq \|e''\| \cdot \|(\phi - \pi_h\phi)''\|$$

$$\leq \|e\|_E$$

$$\|e\|^2 \leq c\|h^2\phi^{(4)}\| \cdot c\|h^2u^{(4)}\|$$

Let $h_j \leq h_{max}$ $\forall j$

$$\|e\|^2 \leq c^2 h_{max}^2 \|\phi^{(4)}\| \cdot \|u^{(4)}\|$$

$$\|e\|^2 \leq c^2 h_{max}^2 \cdot \frac{\|\phi^{(4)}\|}{\|e\|} \cdot \|e\| \cdot \|u^{(4)}\| \quad \left(\frac{\|\phi^{(4)}\|}{\|e\|} = 1\right)$$

$$\Rightarrow \boxed{\|e\| \leq c^2 h_{max}^4 \|u^{(4)}\|}$$

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Abstract Formulation :

So far : 2 examples

(1) $-(au')' = f$ in $(0,1)$

$u(0) = u(1) = 0$

$V = \{v : \int_0^1 [v^2 + (v')^2]\,dx < \infty,\ v(0) = v(1) = 0\}$

$B(u,v) = \int_0^1 au'v'\,dx$

$L(v) = \int_0^1 fv\,dx$

Var Prob : Find $u \in V$ s.t $B(u,v) = L(v)\ \forall\ v \in V$

$F(w) = \frac{1}{2} B(w,w) - L(w)$

Energy Min Prob : Find $u \in V$ s.t $F(u) \le F(w)\ \forall\ w \in V$.

(2) Beam eqn :

$V = \{v : \int_0^1 [v^2 + (v')^2 + (v'')^2]\,dx < \infty,\ v(0) = v(1) = v'(0) = v'(1) = 0\}$

$B(u,v) = \int_0^1 u''v''\,dx$

and $L(v) = \int_0^1 fv\,dx$.

Note : we seek a ~~continuous~~ conforming FEM : $V_h \subset V$.

(non-conforming FEM : $V_h \not\subset V$.)

Abstract formulation : Consider the domain $\Omega \subset \mathbb{R}^d$.

Define (derivatives via a multi-index)

Let $\underline{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_d)$ be a d-tuple, where $\alpha_1, \alpha_2, \dots, \alpha_d$ are non-negative integers

We refer to $\underline{\alpha}$ as a multi-index ; the length of $\underline{\alpha}$ is

$|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_d$.

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The derivative of V is:

$$D^{\alpha} v = \frac{\partial^{|\alpha|} v}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_d^{\alpha_d}}$$

Ex: If $d = 2$ and $|\alpha| = 2$

$$D^{(2,0)} v = \frac{\partial^2 v}{\partial x_1^2}$$

$$D^{(0,2)} v = \frac{\partial^2 v}{\partial x_2^2}$$

$$D^{(1,1)} v = \frac{\partial^2 v}{\partial x_1 \partial x_2}$$

Define: $L^1(\Omega)$ norm:

$$\|u\|_{L^1(\Omega)} := \int_{\Omega} |u| \, dx$$

Defn: $L^1_{loc}(\Omega) := \{ v : v \in L^1(K) \text{ for any compact subset } K \text{ inside } \Omega \}$

Defn: (Sobolev space)

$$W^{r,q}(\Omega) = \{ v \in L^1_{loc}(\Omega) : \|v\|_{W^{r,q}(\Omega)} < \infty \}$$

where

$$\|v\|_{W^{r,q}(\Omega)} := \left[\sum_{|\alpha| \le r} \|D^{\alpha} v\|^q_{L^q(\Omega)} \right]^{1/q}$$

$$1 \le q \le \infty$$

$$r \ge 0$$

and

$$\|v\|^q_{L^q(\Omega)} = \int_{\Omega} |v|^q \, dx$$

Defn: (Sobolev w/ Homogeneous B.Cs)

$$W^{r,q}_0(\Omega) = \{ v \in W^{r,q}(\Omega) : D^{\alpha} v \big|_{\partial\Omega} = 0 \text{ for } |\alpha| < r \}$$

Defn: (Hilbert Space): When $q=2$ we write $W^{r,2}(\Omega) = H^r(\Omega)$. This is called a Hilbert Space.

Norm: $\|v\|_{H^r(\Omega)} = \sqrt{(v,v)_{H^r(\Omega)}}$

Inner Product: $(v,w)_{H^r(\Omega)} := \sum_{|\alpha|\le r} \int_\Omega (D^\alpha v)(D^\alpha w)\, d\underline{x}$

Note: $r=0$ also makes sense: $H^0(\Omega) = L^2(\Omega)$.

Defn: (Semi-norms): $|v|_{H^r(\Omega)} := \sqrt{\sum_{|\alpha|=r} (D^\alpha v)^2 d\underline{x}}$

Note: for the remainder of discussion, Let V be a Hilbert Space w/ inner product $(\cdot,\cdot)_V$ and corresponding norm $\|\cdot\|_V$.

Defn: A mapping $L: V \to \mathbb{R}$ is called a linear functional if $L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$ for all $\alpha, \beta \in \mathbb{R}$ and all $u, v \in V$. We say that L is bounded in the norm $\|\cdot\|_V$ if there exists a consistent $\tilde{L} > 0$ such that

$$|L(v)| \le \tilde{L} \|v\|_V \quad \forall\, v \in V.$$

Note: A bounded linear functional is continous:

$$L(v) - L(w) = L(v-w)$$

$$\Rightarrow |L(v) - L(w)| = |L(v-w)| \le \tilde{L}\|v-w\|_V \quad \forall\, v, w \in V.$$

The converse is also true: a continous linear functional is bounded.

Defn: The set of bounded linear functionals on V is termed the dual space of V. and is denoted by V'. V' is equipped w/ a norm.

$$\|L\|_{V'} = \sup_{\substack{v \in V \\ v \neq 0}} \frac{L(v)}{\|v\|_V} \le \tilde{L}$$

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ex: (Dual space of $L^q(\Omega)$) Consider the linear functional

$$L(v) = \int_\Omega f(x)\, v(x)\, dx \quad \text{for all } f \in L^{q'}(\Omega) \text{ where } \frac{1}{q} + \frac{1}{q'} = 1.$$

Hölder's Inequality: $\|uv\|_{L^1(\Omega)} \leq \|u\|_{L^q(\Omega)} \cdot \|v\|_{L^{q'}(\Omega)}$

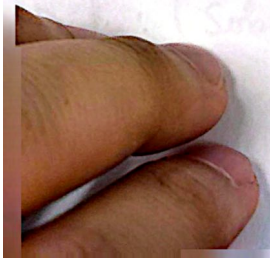
$\forall\ u \in L^q(\Omega)$ & $v \in L^{q'}(\Omega)$

$\therefore |L(v)| \leq \|f\|_{L^{q'}(\Omega)} \|v\|_{L^q(\Omega)}$

$\therefore$ every function $f \in L^{q'}(\Omega)$ can be viewed as a linear bounded functional on $L^q(\Omega)$

Every bounded linear functional on $L^q(\Omega)$ is of this form,

$\therefore$ $L^{q'}(\Omega)$ can be viewed as the dual space of $L^q(\Omega)$.



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